

When should multi-agent **LLM** teams stop?

First systematic study of termination dynamics across 14 coordination topologies.
2,200+ unified runs · 7 experiments · 5 LLMs · COLM 2026 target.

14

TOPOLOGIES

2,200+

RUNS

7

EXPERIMENTS

5

LLMS

Termination Regret

Multi-agent LLM systems (AutoGen, LangGraph, CrewAI) commonly use a fixed `max_turns=10` across all topologies. But a 2-agent chain and a 4-agent debate have [fundamentally different convergence behaviors](#). Applying a uniform stop rule produces what we term **termination regret** — three failure modes:

Over-termination

Debate team continues 4 rounds past consensus, burning tokens with zero quality gain.

Under-termination

Sequential pipeline stops after the draft stage, never reaching the refinement that would have caught factual errors.

Topology mismatch

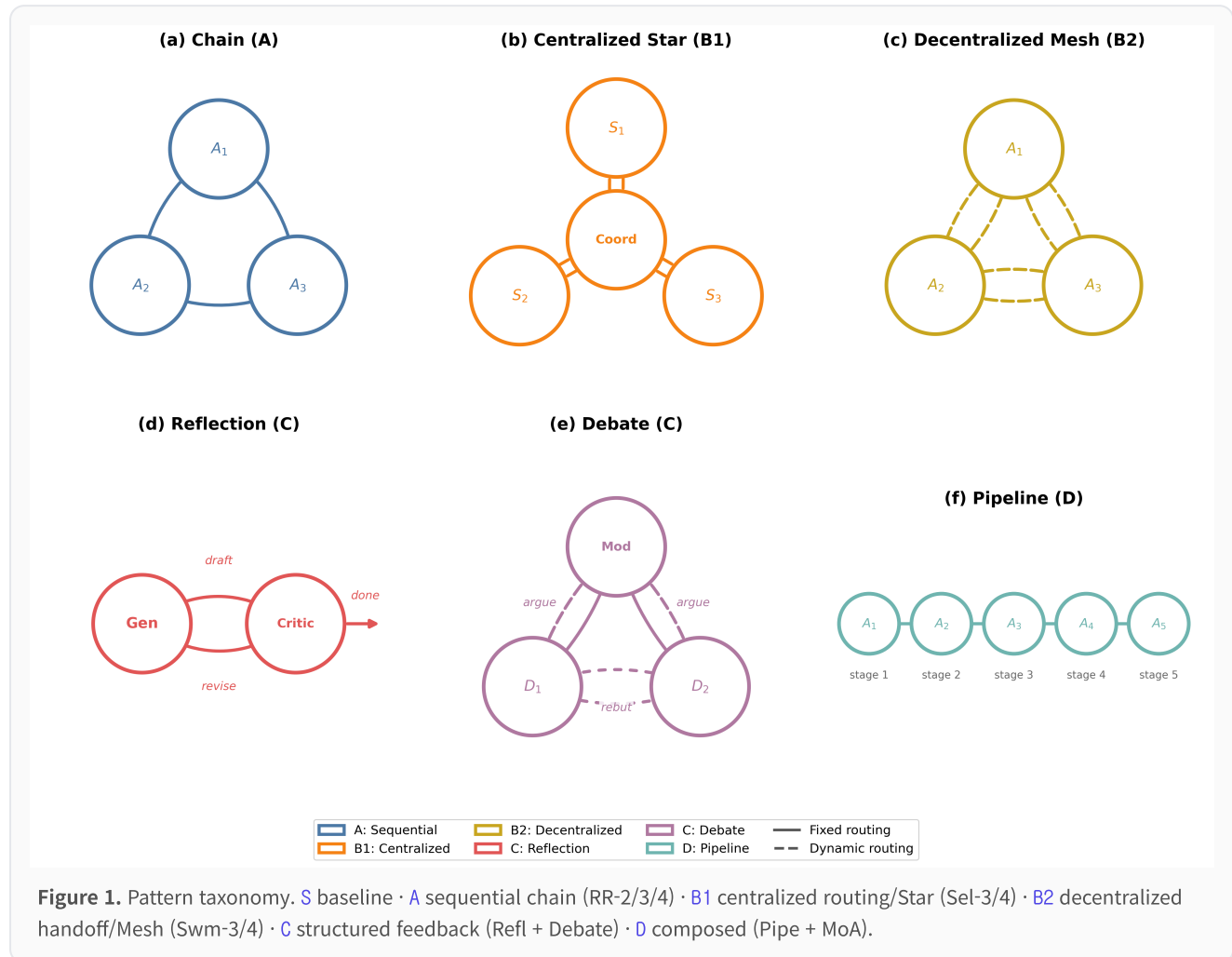
Selector-based team applies max-turn calibrated for flat patterns, cutting routing agent off mid-deliberation.

RESEARCH GAP — PRIOR WORK IN ISOLATION

STUDY	PATTERNS	TOPOLOGY	TERMINATION?
REFRAIN (2025)	1	single-agent CoT	yes
Hu et al. NeurIPS 2025	1	debate-only	yes
Aegean (2025)	—	parallel consensus	yes
MAST (Cemri 2025)	~5	mixed	no
Ours	14	6 categories unified	yes (topology-aware)

14 Coordination Topologies · 6 Categories

Patterns grounded in **Masterman et al. (2025)** Chain/Star/Mesh + **Tran et al. (2025)** centralized/distributed split. The B1/B2 decomposition is a contribution of this work.



Coordination Architectures

Figure 2: Message Flow Architectures for Each Pattern Category

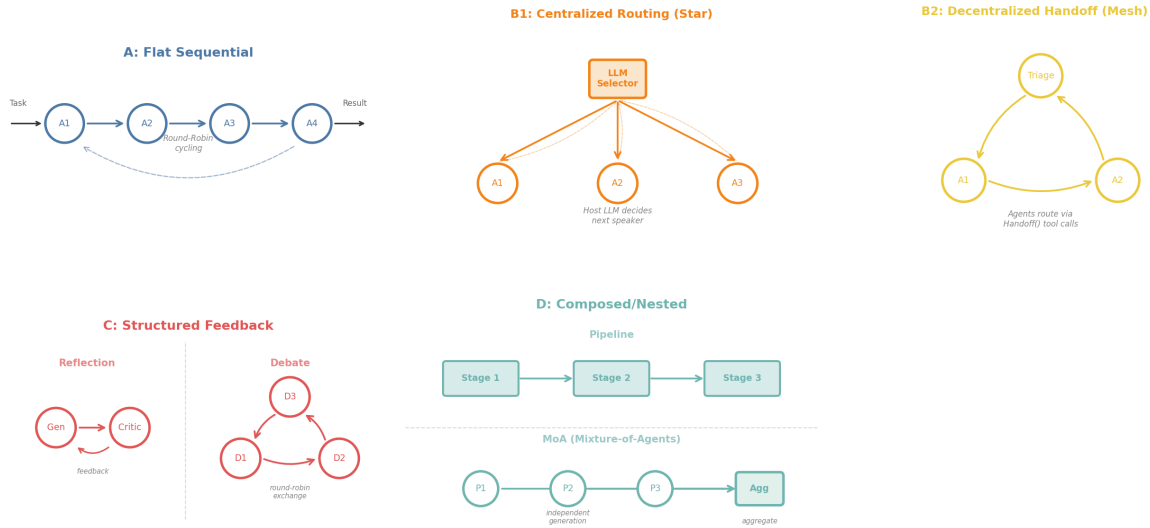


Figure 2. Communication patterns per category. Chain passes context linearly; Star concentrates routing in a selector node; Mesh enables peer-to-peer handoff; Reflexion adds feedback loops; Debate parallelizes argument; Composed combines primitives.

Why categories matter: previous literature treated B (dynamic routing) as one bucket. Our experiments show **centralized (B1) and decentralized (B2) scale opposite directions** — sub-linear vs super-linear. New cost hierarchy: $A \approx B2 < B1 \approx C \ll D$.

Experimental Design

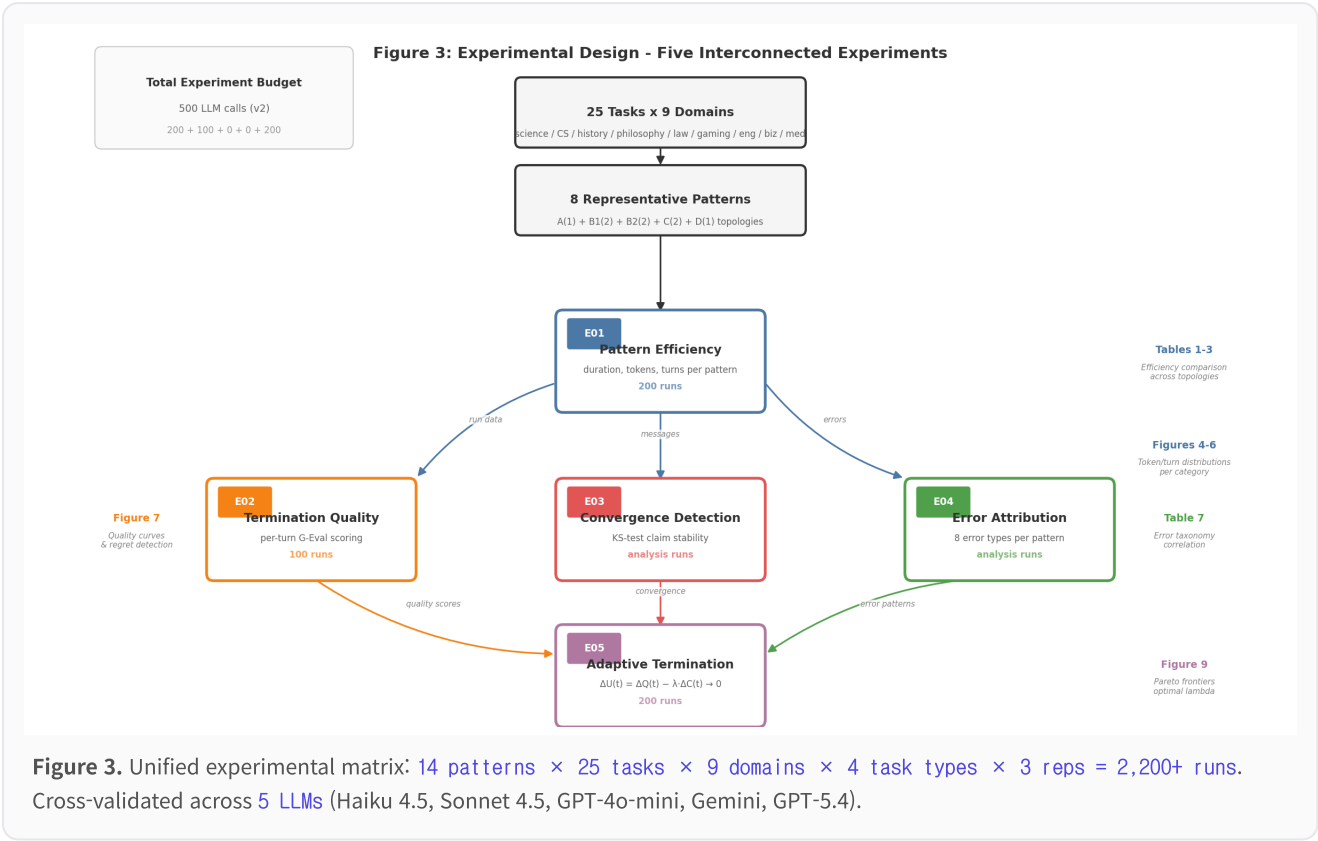


Figure 3. Unified experimental matrix: 14 patterns × 25 tasks × 9 domains × 4 task types × 3 reps = 2,200+ runs. Cross-validated across 5 LLMs (Haiku 4.5, Sonnet 4.5, GPT-4o-mini, Gemini, GPT-5.4).

Tasks

25 tasks across 9 domains (science, CS, history, philosophy, law, games, engineering, business, medicine) × 4 types (factual, analytical, technical, creative).

Models

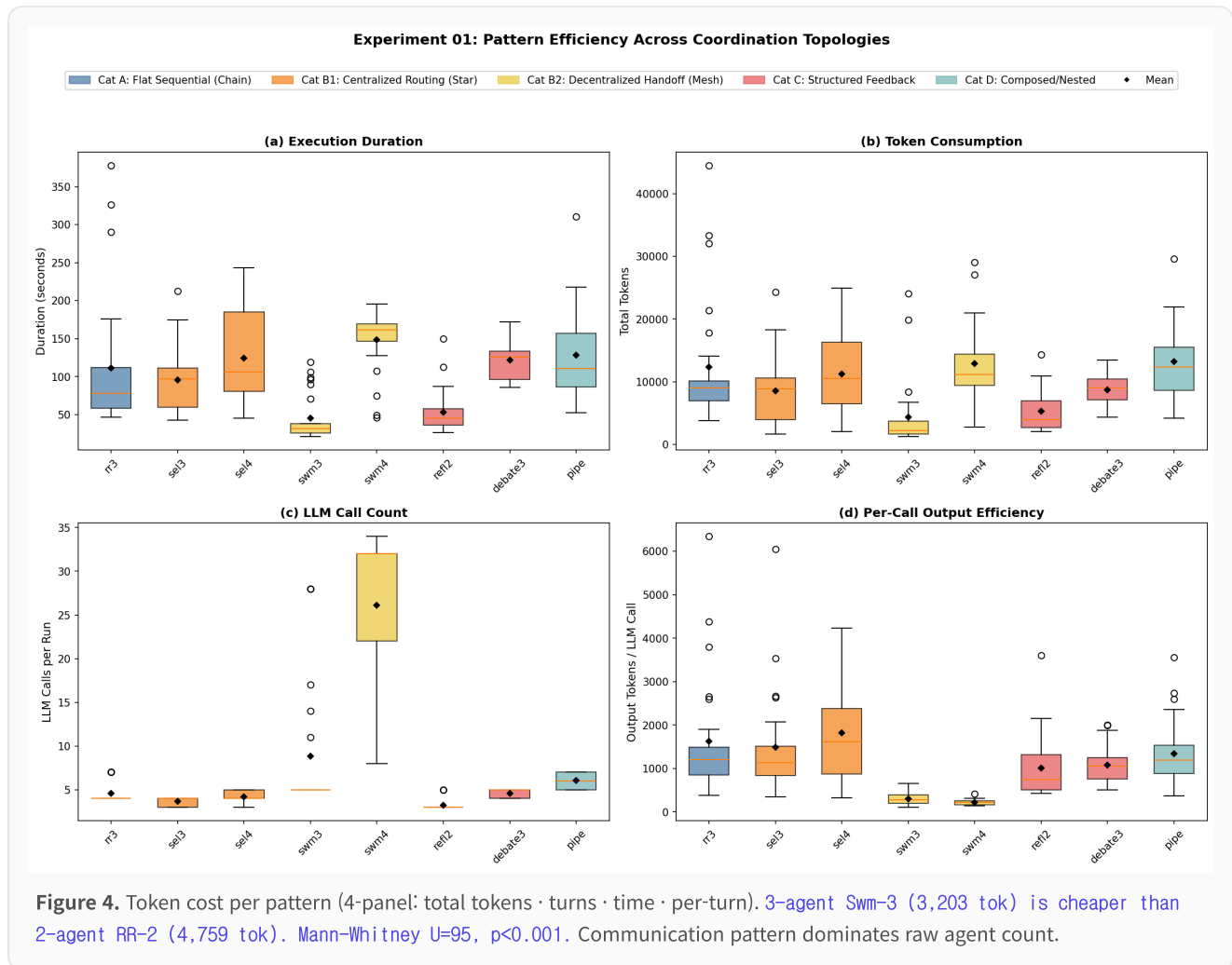
Claude Haiku 4.5 (agents) · Sonnet 4.5 (G-Eval judge) · GPT-4o-mini (cross-val p=0.900) · Gemini · GPT-5.4.

Statistics

Kruskal-Wallis · Mann-Whitney U · Cohen's d · 5-fold CV · Spearman ρ · bootstrap CI.

F1 — Cost Inversion

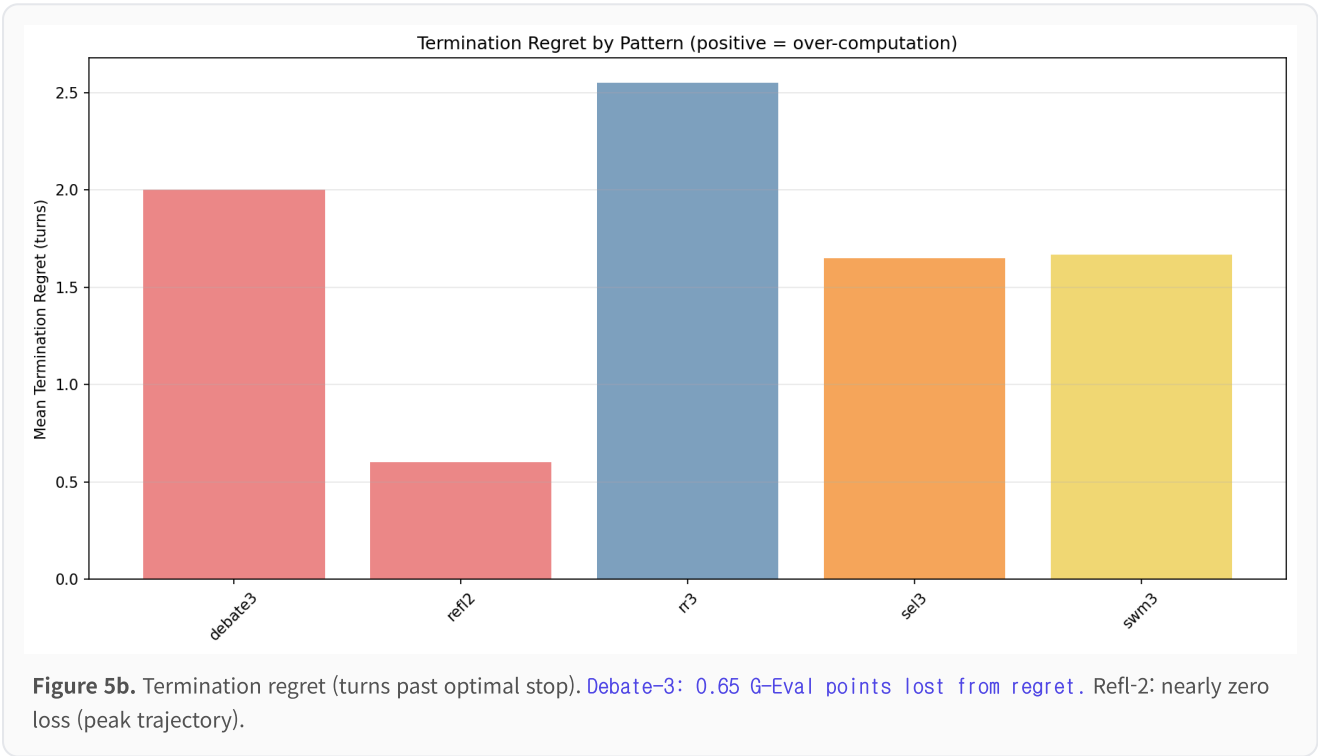
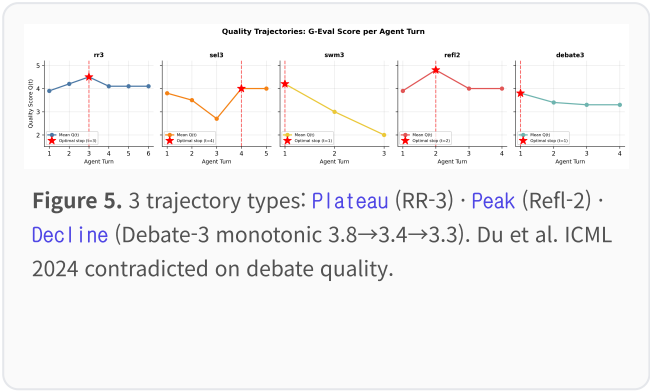
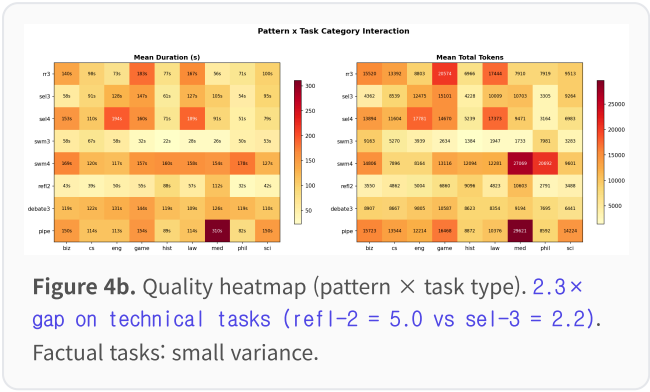
Exp01 · 980 runs



Solo (1,287) < Swm-3 < RR-2 < Ref1-2 < Sel-3 < Deb-3 < RR-3 < Sel-4 ≈ Deb-4 < Swm-4 < Pipe < RR-4 < MoA (22,333) • **17.4×**
cost gap

F2 — Quality by Task & Trajectory

Exp02 · G-Eval



F3 — Semantic Convergence

Exp03 · Sentence-BERT

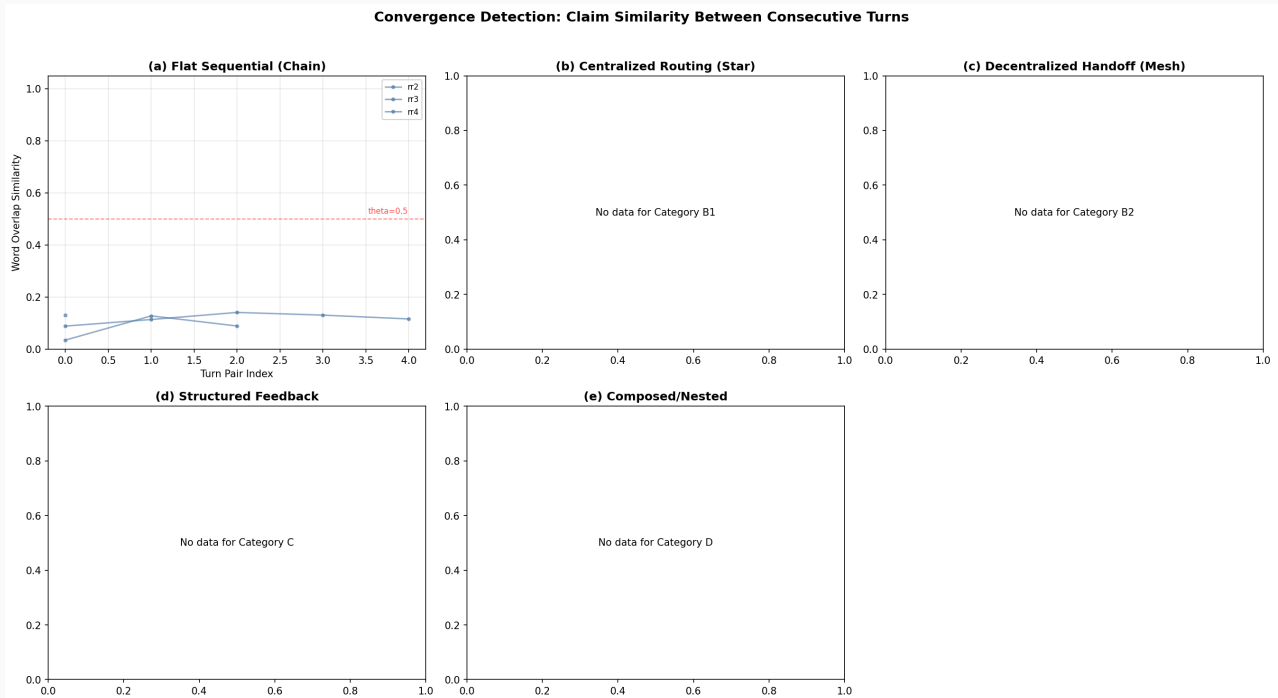
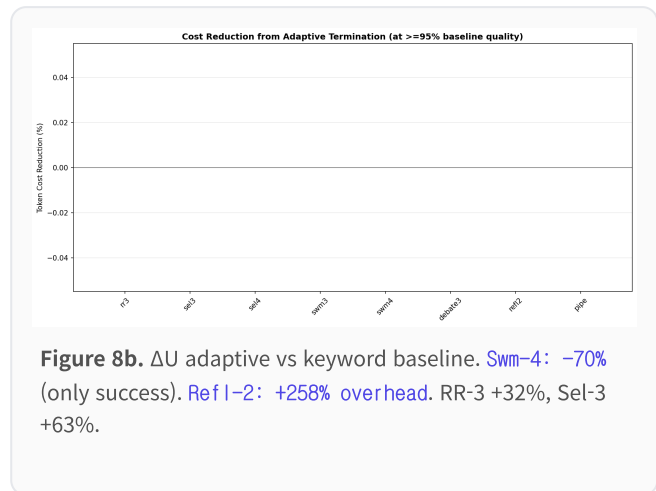
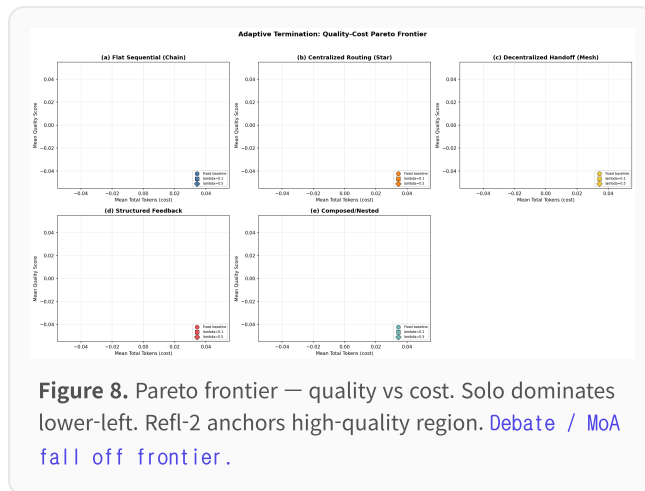


Figure 6. Semantic stability per turn (Sentence-BERT cosine). Pipe 32% > B1 26% > A 16% > B2 12% > C 0%. Debate-3 has highest cosine (0.673) but never converges — every turn introduces new arguments. Convergence signals are [topology-dependent](#); one-size-fits-all detection fails.

Engineering takeaway: Convergence-based termination only viable for Pipeline / B1 / A. For B2 / Debate, use [topology-specific signals](#) (turn loop detection, argument novelty score).

F4 — ΔU Negative Result

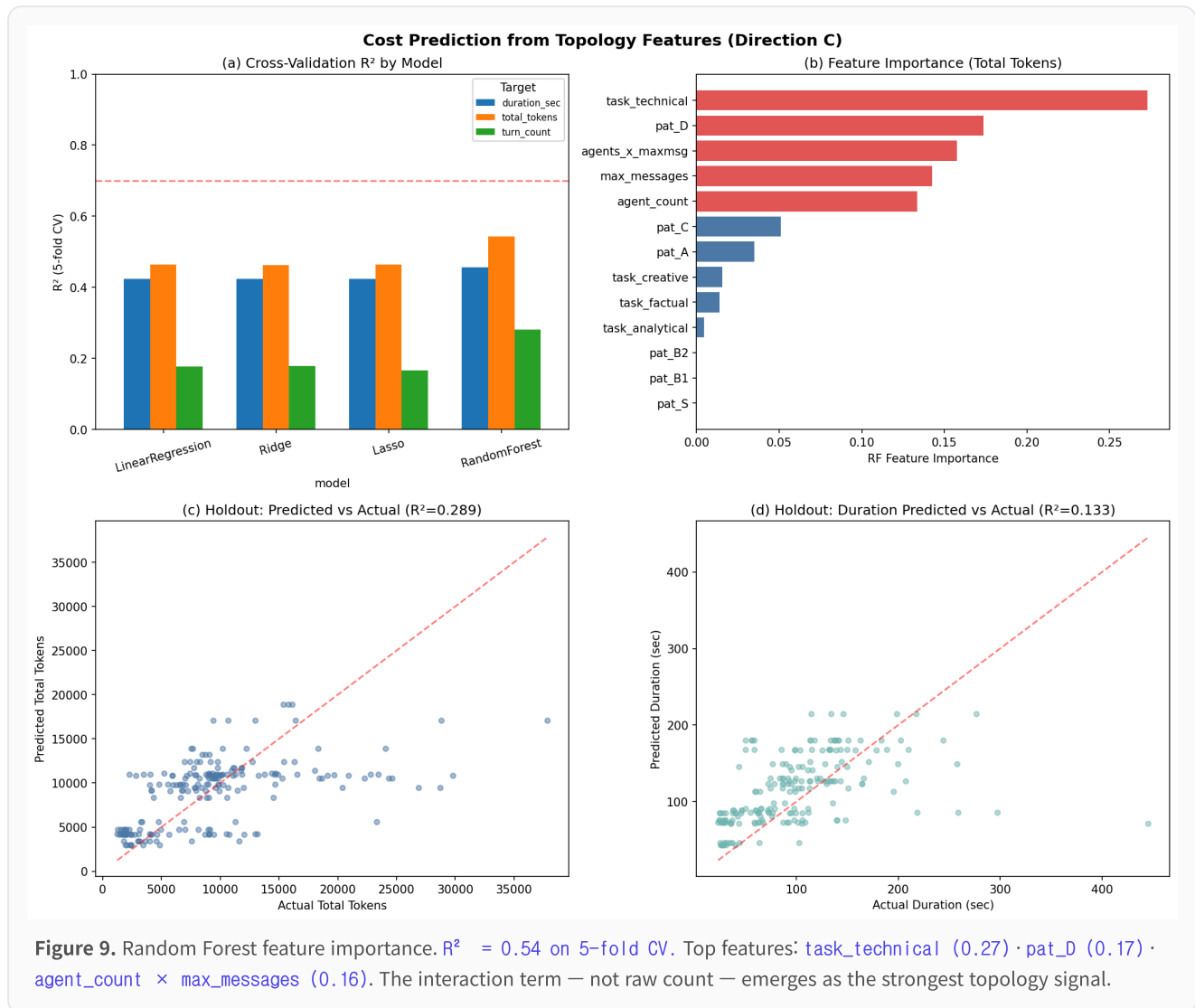
Exp05 · 800 runs



Negative result with practical value: 7 of 8 patterns: keyword TERMINATE is near-optimal. $\Delta U(t) = \Delta Q(t) - \lambda \cdot \Delta C(t)$ is a diagnostic lens, not a replacement. Hybrid strategy: keyword everywhere, ΔU only for B2 (decentralized handoff) where keyword fails (28% success rate).

F5 — Cost Prediction Model

Exp06 · 718 runs · 5-fold CV



Honest boundary: Topology features alone explain 54% of cost variance. The remaining 46% is task-content complexity — not predictable from coordination structure. Holdout on Exp05 task suite: $R^2=0.29$ (recalibration needed across task domains).

F6 — Difficulty Dominates Pattern

Exp07 · 669 runs · 5 LLMs

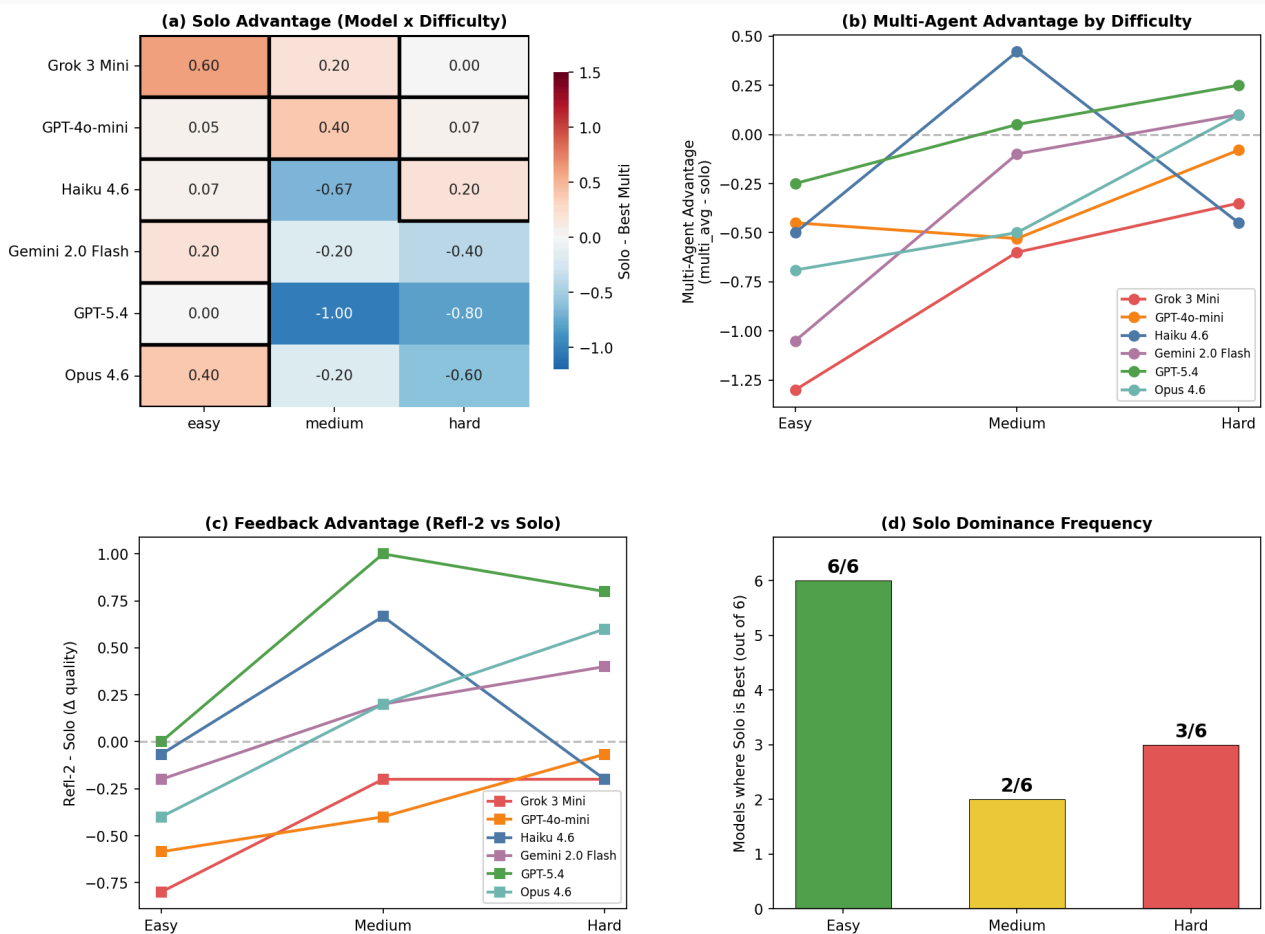


Figure 10. Quality by difficulty × pattern across 5 models. **Difficulty effect** $\eta^2 = 0.363$ (large) vs **pattern effect** $\eta^2 = 0.039$ — $9.3\times$ difference. Solo dominates Easy & Hard. Refl-2 helps only Haiku & GPT-5.4 at Medium (Cohen's $d=1.26$ / 1.83).

Recommendation matrix: Solo as default for cost-efficiency. Activate multi-agent coordination only when task difficulty approaches the model's capability frontier. The previously-reported inverted-U pattern holds for only 1 of 5 models (Haiku). **Capability-aware topology selection** beats blanket multi-agent strategies.

Seven Contributions

1 · Cross-topology study

First systematic comparison of 13 patterns × 7 experiments × 2,200+ runs in unified AutoGen framework. Prior work: 1-3 patterns max.

2 · Termination regret metric

Post-hoc G-Eval trajectory analysis identifies optimal stop point vs actual stop point. Quantifies systematic over-/under-computation.

3 · Claim-level convergence

Sentence-BERT semantic stability extending Hu NeurIPS 2025 KS-test to all 13 topologies. Shows convergence is topology-dependent.

4 · Pattern-specific error taxonomy

Aligned with MAST (Cemri 2025): topology-correlated error types (token overflow / turn loop / missing keyword) per pattern.

5 · ΔU diagnostic framework

Negative result: 7/8 keyword optimal. Positive result: B2 -70% with ΔU . Marginal utility as [principled diagnostic lens](#), not optimizer.

6 · Cost prediction model

[RF \$R^2 = 0.54\$](#) · agents × max_messages strongest predictor · 46% task-content boundary acknowledged.

7 · Difficulty-aware recommendation matrix

5 LLMs × 3 difficulty levels mapping. [Solo dominates cost-efficiency at every level](#). Refl-2 helps only weak models at medium difficulty (capability moderation).

"Termination strategy should be a first-class design decision."

Framework released — 13 pattern implementations · 7 experiment runners · analysis scripts.